

>_ CTerm Blueprint v2.0

Ultra-Lightweight, Zero-Knowledge E2EE, Local-First SSH Connection Manager

Author: Cecep Azhar • License: MIT (Core Desktop Engine) • Architecture: Tauri 2.0 / Rust Core + Svelte 5 / TailwindCSS

Having used Termius and traditional SSH GUI tools for years across multi-device and multi-platform environments, several critical flaws degrade professional developer experience:

- **Extreme Memory Bloat:** Electron-based SSH clients consume 400MB to 1GB+ RAM per instance, draining laptop battery and lagging on lower-spec hardware.
- **Vendor Lock-in & Data Revocation:** The moment a subscription ends, access to cloud configurations, host groups, custom snippets, and sync preferences is revoked or wiped out.
- **Incomplete Configuration Backup:** Local exports fail to capture the entire environment context (known hosts, custom key bindings, environment variables, AI settings).
- **Privacy & Telemetry Risks:** Kredensial dan metadata tersimpan di server pihak ketiga dengan enkripsi proprietary yang tidak terverifikasi independen.

- CTerm Core Philosophy (5 Pillars):**
- 1. **Very Light:** Tauri 2.0 (Rust) + System Webview → **< 35MB RAM** usage, **< 8MB binary**.
 - 2. **Stable:** SQLite WAL Mode + Native Rust Async PTY Engine (`russh`). Zero panic, 100% thread-safe.
 - 3. **Privacy First:** Zero-Knowledge E2EE (Argon2id + AES-256-GCM). Key lokal tidak pernah keluar tanpa enkripsi.
 - 4. **Collaboration Next:** Asymmetric E2EE Team Vault sharing via ECDH (Curve25519) + \$1/month team seats.
 - 5. **Sync Next:** Self-Hostable or Managed Sync Cloud (\$1 month 1, \$3/mo recurring). **Jika langganan habis, DATA LOKAL TETAP 100% UTUH & BERFUNGSI SELAMANYA!**

Module / Feature	Status in v2 Open-Source App	Technical & Functional Realization
Hosts & Groups Management	ACTIVE LOKAL	Organisasi hierarkis host SSH dengan dukungan tag, grup teratur, variasi port/user, password/private key terenkrpsi SQLite. Filter pencarian instan.
Interactive PTY Terminal	ACTIVE LOKAL	Mesin PTY berbasis xterm.js v5 + Rust `russh` tunnel. Mendukung 256-color, truecolor, tab management, split pane horizontal/vertikal, dan bracketed paste.
TOFU Anti-	ACTIVE LOKAL	Trust-On-First-Use Host Key Verification. Fingerprint server

Module / Feature	Status in v2 Open-Source App	Technical & Functional Realization
MITM Verification		dicatat persisten di tabel SQLite `host_keys`. Mencegah serangan Man-In-The-Middle.
Files Manager (SFTP)	ACTIVE LOKAL	Integrasi modul SFTP native Rust. File explorer dua panel (lokal vs remote), drag-and-drop upload/download, resume transfer, dan permission editor.
Snippets Engine	ACTIVE LOKAL	Perpustakaan skrip CLI dengan variabel dinamis (`{{host}}`, `{{user}}`). Fitur Auto-Enter atau inspeksi sebelum dieksekusi ke terminal aktif.
Port Forwarding (Tunnels)	ACTIVE LOKAL	Pengelola tunneling SSH Local, Remote, dan Dynamic (SOCKS5 Proxy). Status indikator aktif/nonaktif dalam sekali klik.
Server Real-time Monitoring	ACTIVE LOKAL	Grafik statistik server (CPU, RAM, Disk Usage, Network Traffic, System Load) secara real-time via SSH background stream tanpa perlu install agent tambahan.
Command Audit Logs	ACTIVE LOKAL	Pencatatan riwayat perintah dan output terminal secara terpusat untuk audit keamanan. Terpotong otomatis 5KB untuk menghemat ruang SQLite.
SSH Keys Manager	ACTIVE LOKAL	Generator kunci SSH lokal (RSA 4096, Ed25519), import/export private key, serta copy public key otomatis ke server (`ssh-copy-id` internal).
AI Terminal Copilot	ACTIVE LOKAL	Asisten AI kontekstual untuk penjelasan komando CLI, analisis log error, dan pembuat skrip. Konfigurasi provider kustom terenkripsi DEK.
E2EE Multi-Device Sync	DISABLED / NEXT FEATURE	Sinkronisasi enkripsi SQLite antar perangkat via FathCloud / Self-hosted Sync Relay. Tombol UI menampilkan badge <i>"Coming Soon / Managed Cloud"</i> .
Team Vault Collaboration	DISABLED / NEXT FEATURE	Berbagi kredensial host ke anggota tim via ECDH Key Exchange. Tombol UI dalam keadaan ter kunci / siap diaktifkan setelah infra Cloud diluncurkan.

Untuk mempertahankan integritas filosofi Open-Source dan kebebasan pengguna, seluruh fitur **Multi-Device**

Sync dan **Team Collaboration** di dalam repositori publik CATerm v2 akan dipasang dalam kondisi **DISABLED** /

NEXT FEATURE (Mode Placeholder UI Cantik).

- **Trial Onboarding: \$1** untuk bulan pertama (Memvalidasi pengguna serius dan menutupi biaya verifikasi

payment gateway).

- **Managed Cloud Individual: \$3 / bulan** berikutnya (Harga sangat terjangkau dibanding Termius \$10+/bln).

- **Team Collaboration Seats: +\$1 / bulan per anggota tim tambahan.**

- **Infrastruktur Core:** Node backend Microservice (Go/Rust) yang dideploy di atas ****Coolify / FathCloud****

(Hostinger VPS & DigitalOcean).

- **Zero-Knowledge Storage Engine:** Server sync HANYA menyimpan *encrypted SQLite blob* (AES-256-

GCM). Server TIDAK PERNAH memiliki kunci enkripsi (Master Password / DEK).

- **Asynchronous Team Sharing (ECDH):** Setiap pengguna memiliki pasangan kunci publik/privat

Curve25519. Kredensial grup server dienkripsi menggunakan *shared secret* ECDH anggota tim sebelum

diunggah.

- **Self-Host Option vs Managed Official Cloud:** Kode server backend sync juga akan dibuat *lightweight*

(Docker Container) sehingga komunitas dapat melakukan *self-hosting* gratis, sementara pengguna biasa

dapat berlangganan Official Managed Cloud seharga \$3/bln.

```
[Frontend: Svelte 5 + Tailwind + xterm.js v5] | (Tauri 2.0 IPC Commands & Events) [Backend: Rust Core Application]
└─ src/vault/ -> Argon2id + AES-256-GCM Encrypted SQLite (rusqlite) └─ src/ssh/ -> russh PTY Stream & SFTP
Subsystem └─ src/tofu/ -> Fingerprint Persistence & MITM Guard └─ src/ai/ -> Encrypted API Client & Prompt
Formatters └─ src/sync/ -> CRDT LWW-Element-Set Sync Protocol (Disabled in Core)
```